

Visualization through human eyes

Readings for today

- Franconeri, S., L. Padilla, P. Shah, J. Zacks, & J. Hullman (2021). The science of visual data communication: What works. *Psychological Science in the Public Interest*.

Topics

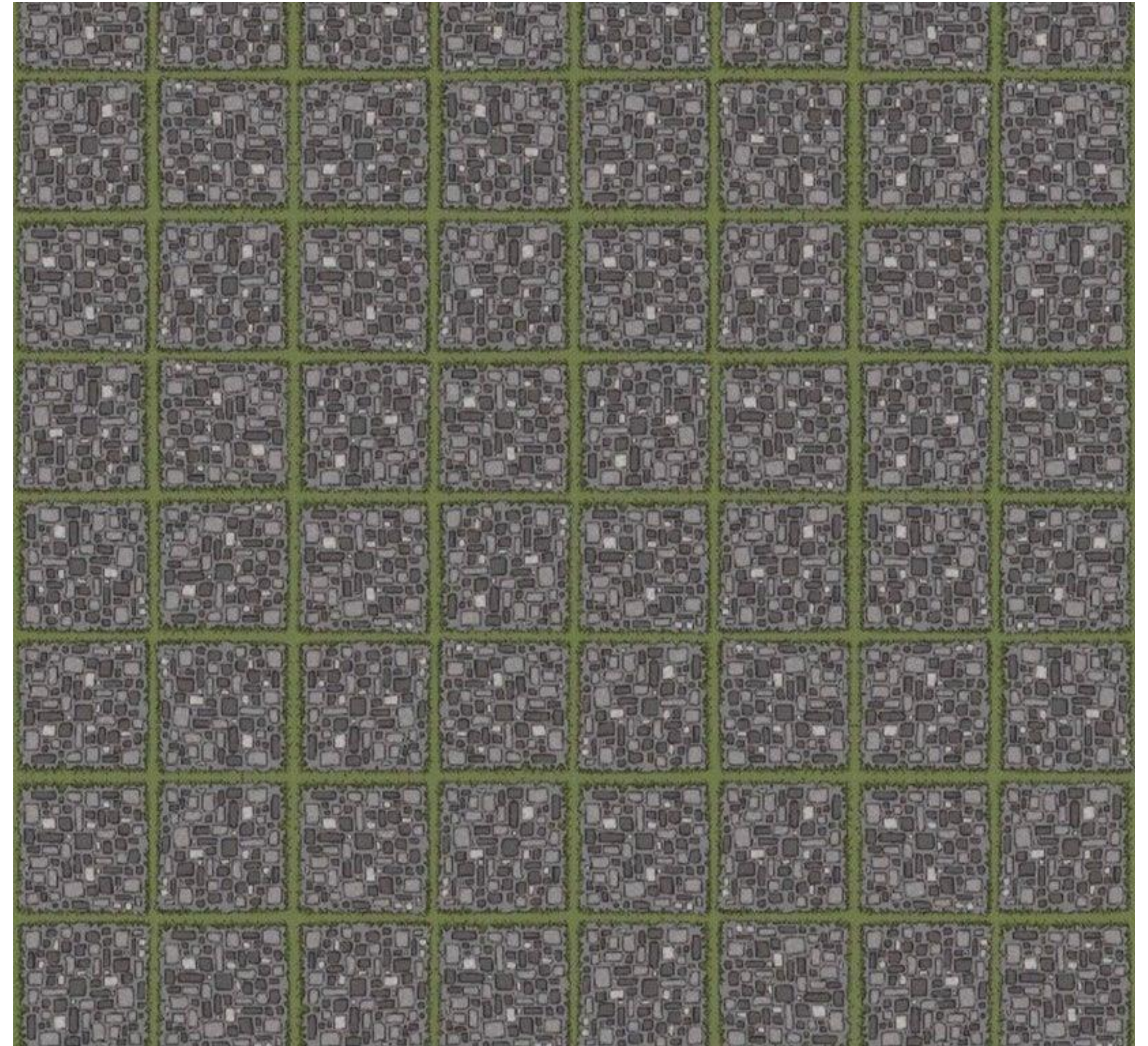
1. What the human visual system “sees”
2. Working with visual biases

What the human visual system “sees”

Seeing is not believing

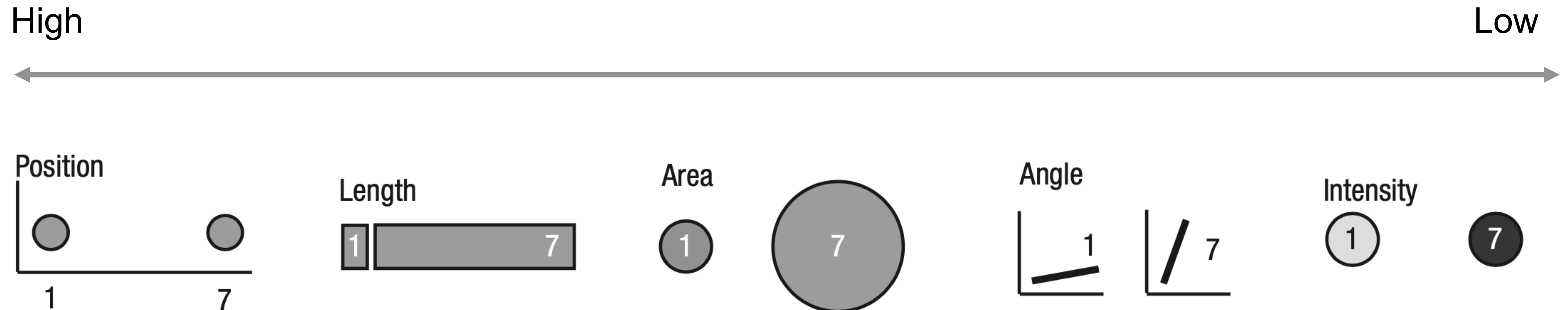
Humans see the world biased by their internal expectations (i.e., biases/priors), not based purely on the senses themselves.

There are no curved lines



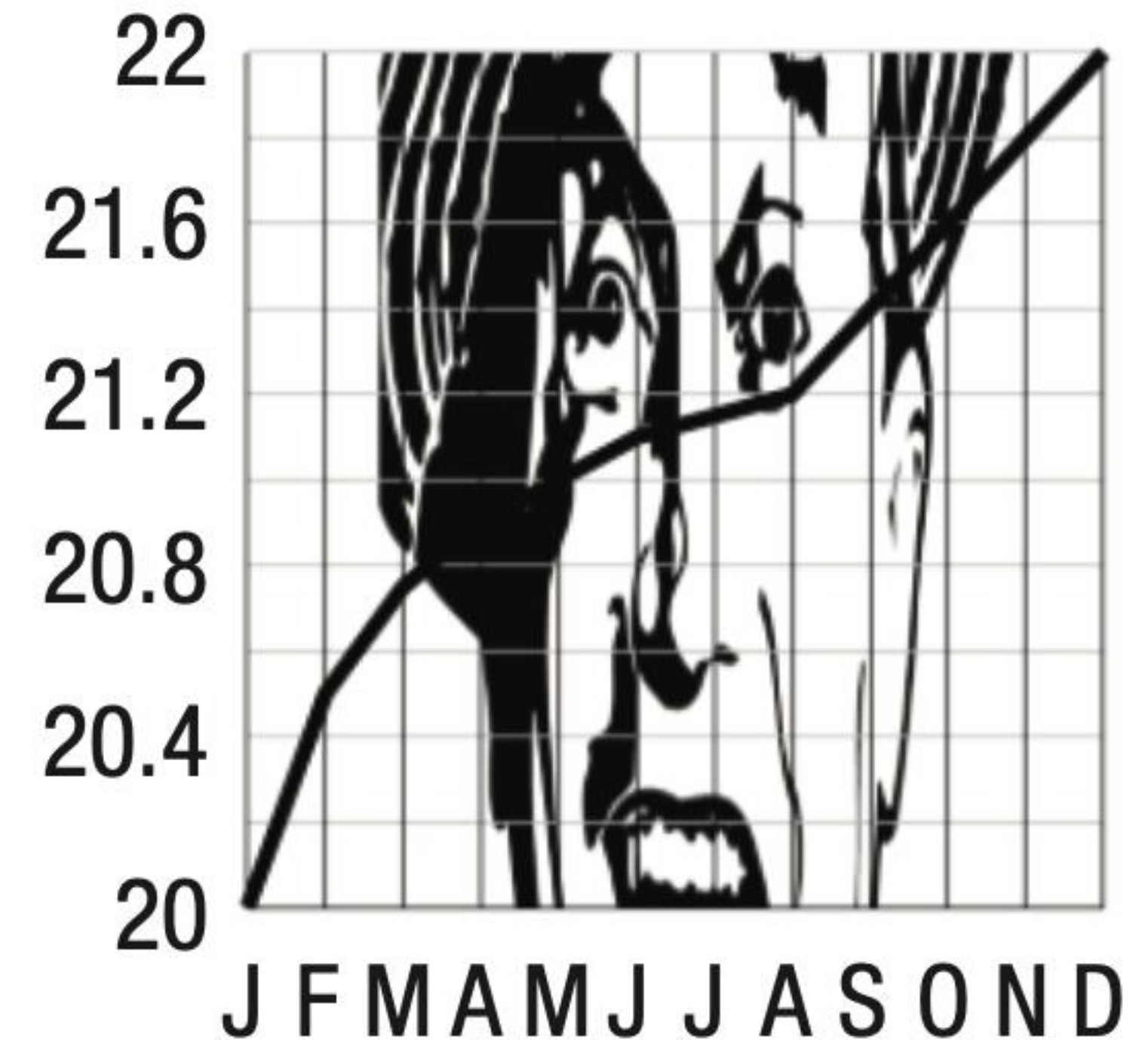
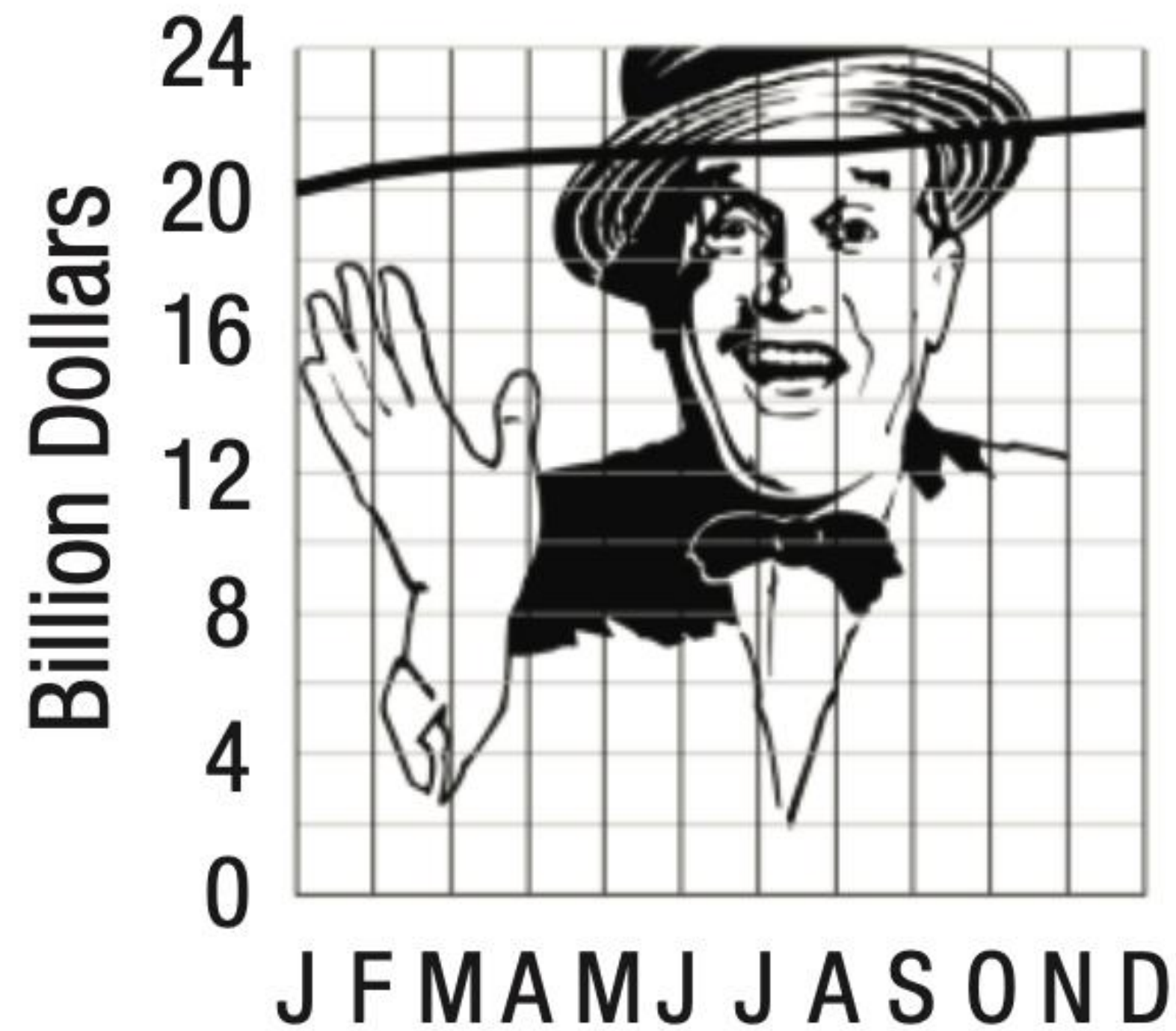
Precision of visual channels

Precision: The ability by which viewers can verbally state the relations between 2 values when visually presented.



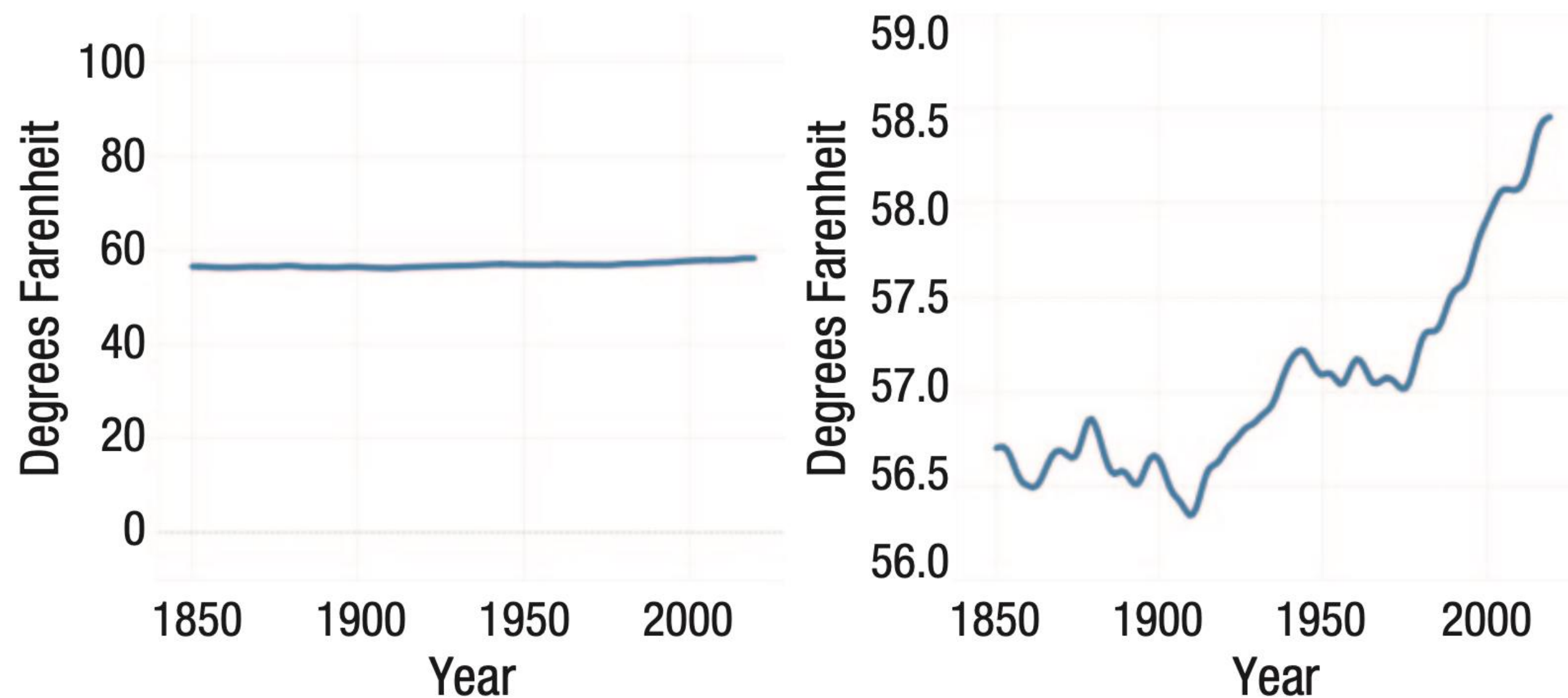
Reference frames → understanding

Mapping visual ratios to numbers can lead to perceptual & inferential errors.



Reference frames → understanding

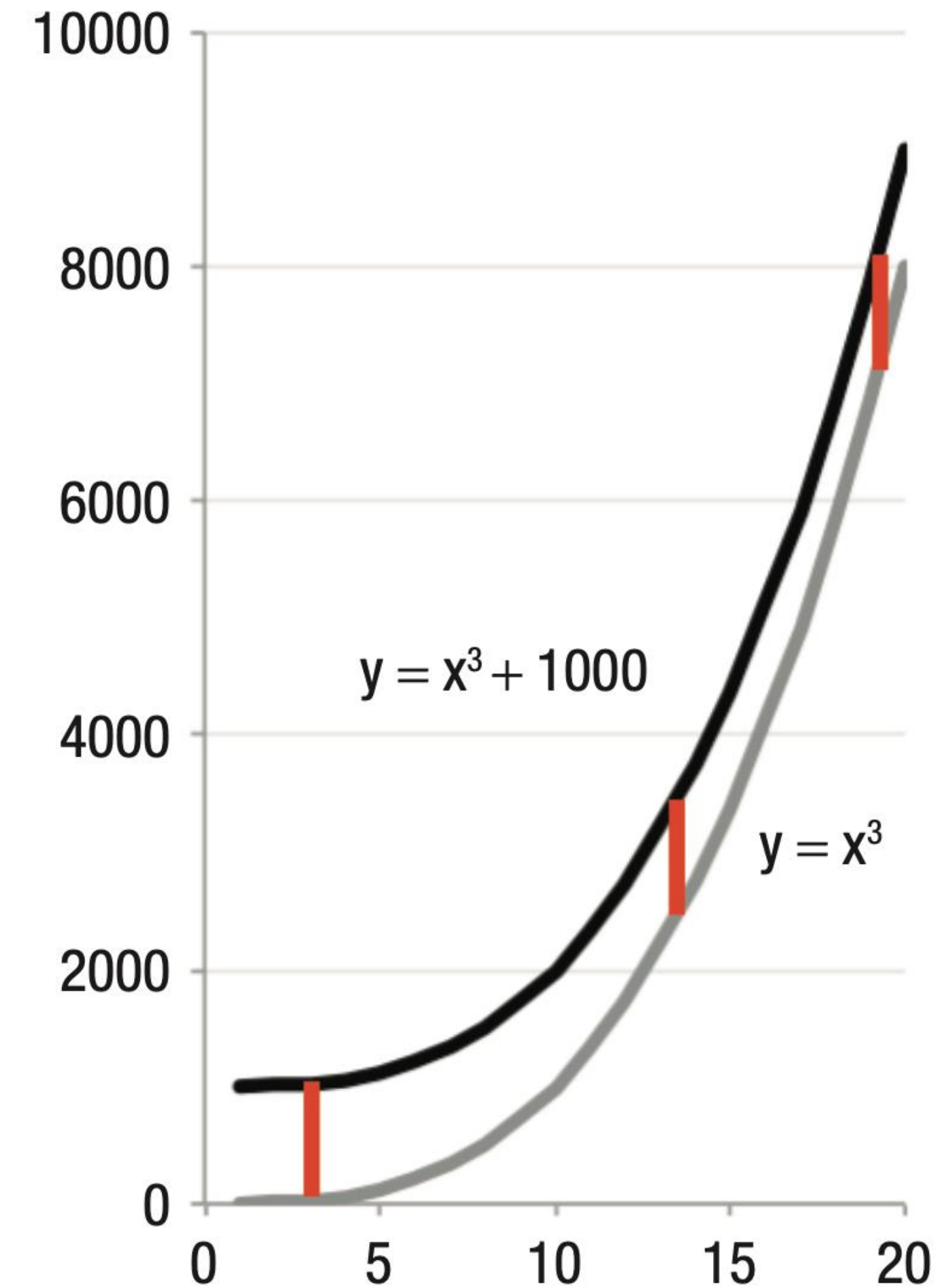
Mapping visual ratios to numbers can lead to perceptual & inferential errors.



Designers should consider the practical relevance of scale, reference frame, etc. when thinking of how they want to the viewer to encode the information.

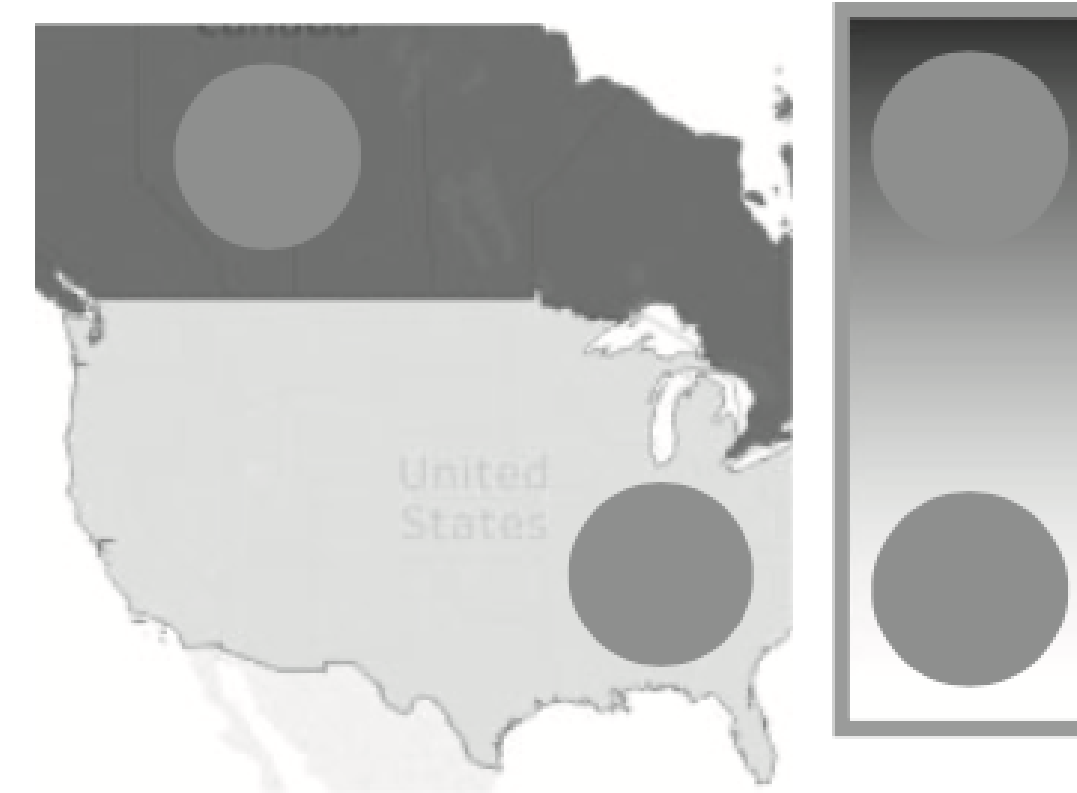
Common illusions & misperceptions

- Hue & position lead to perceived distance/difference errors.



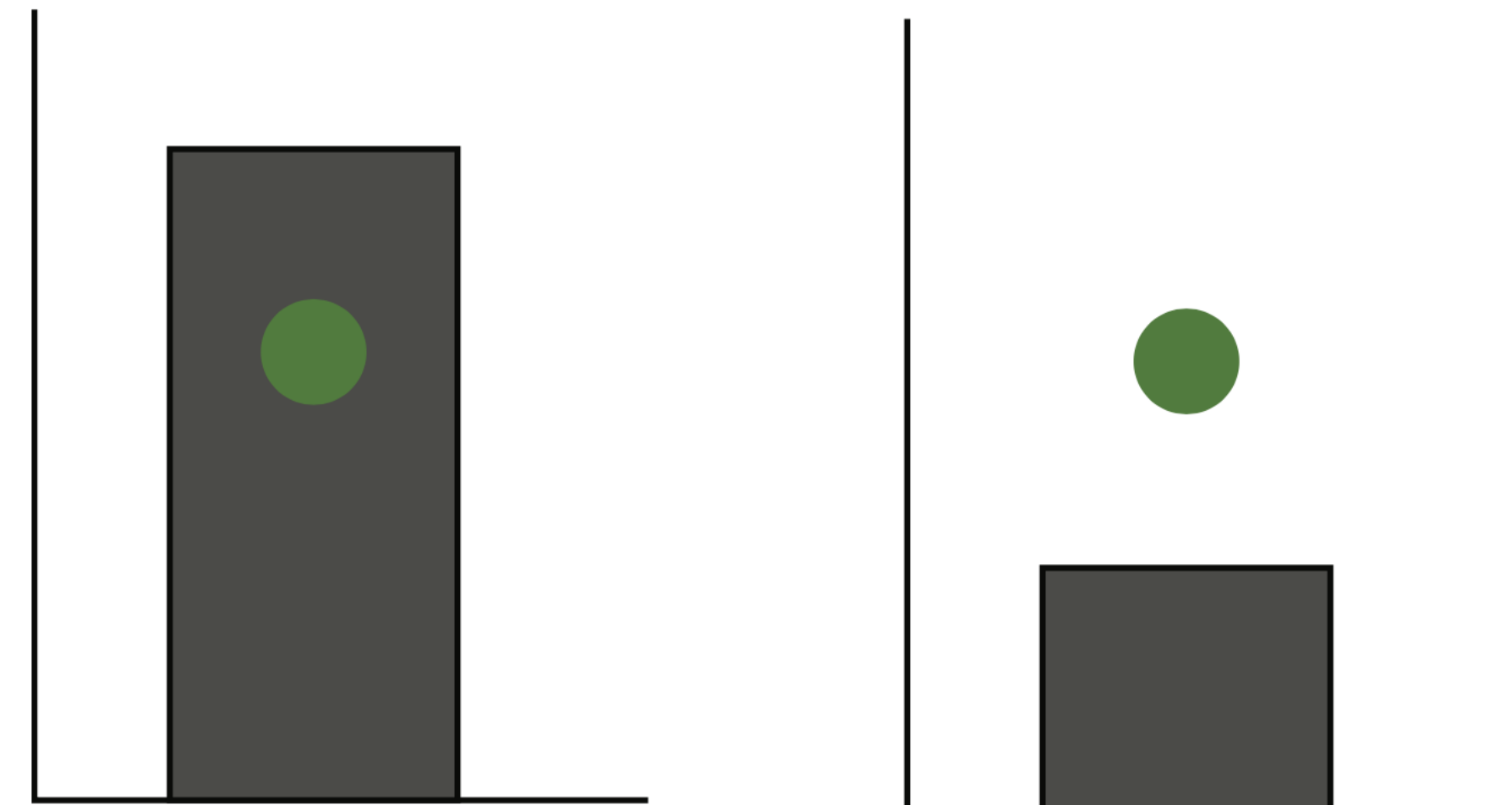
Common illusions & misperceptions

- Hue & position lead to perceived distance/difference errors.
- Contrast & context skew perceptions of intensity.



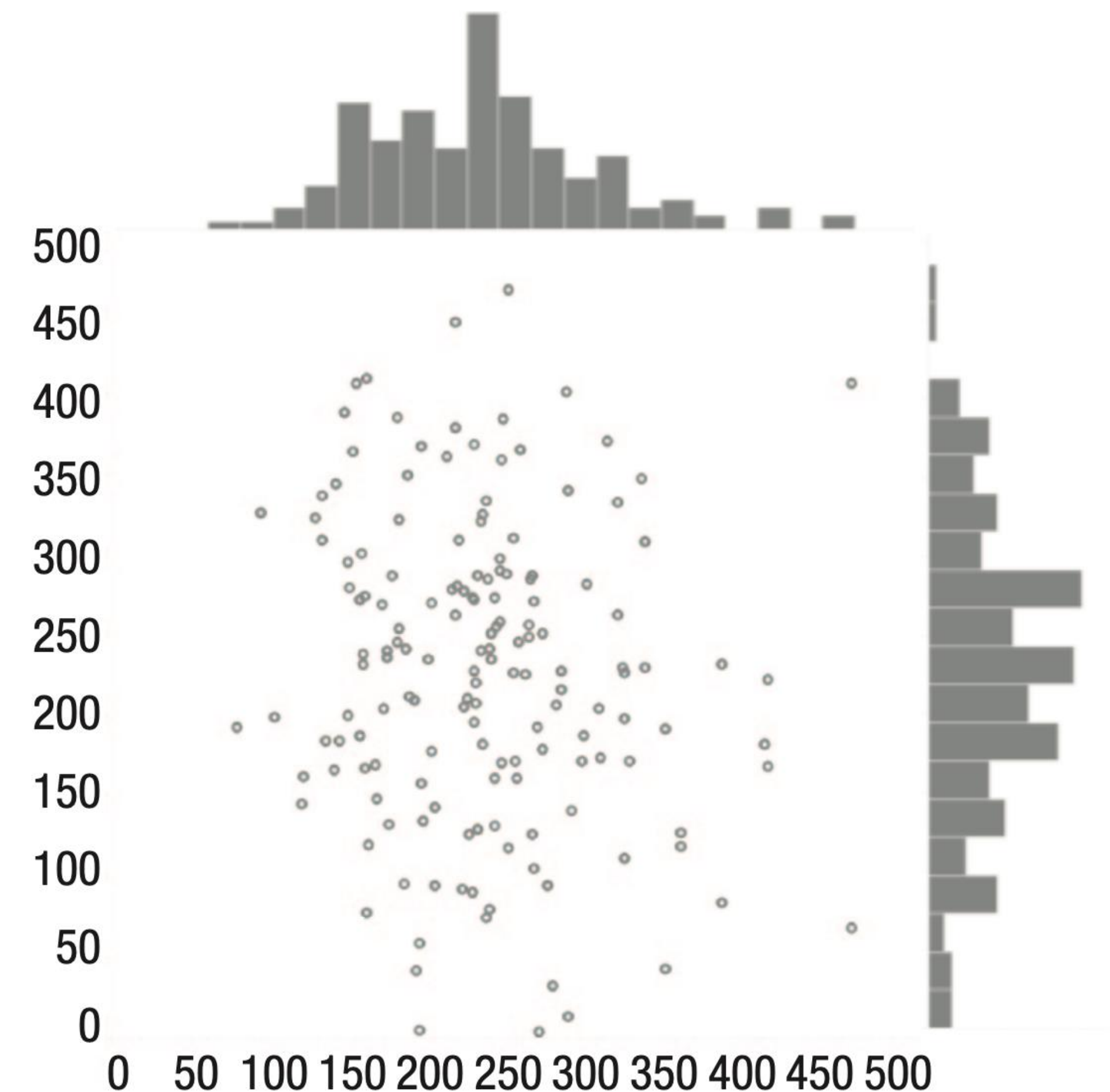
Intensity values can look different depending their backgrounds.

Do not plot intensities on intensities.



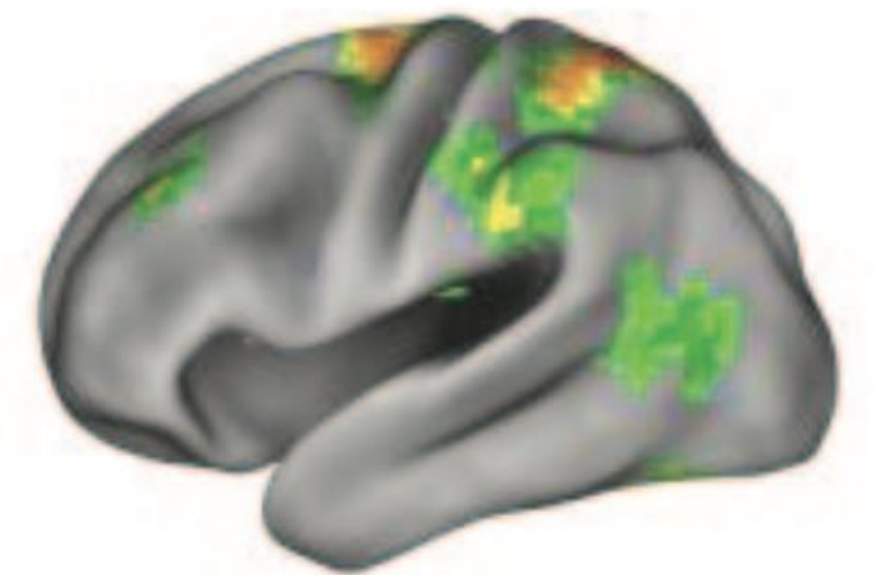
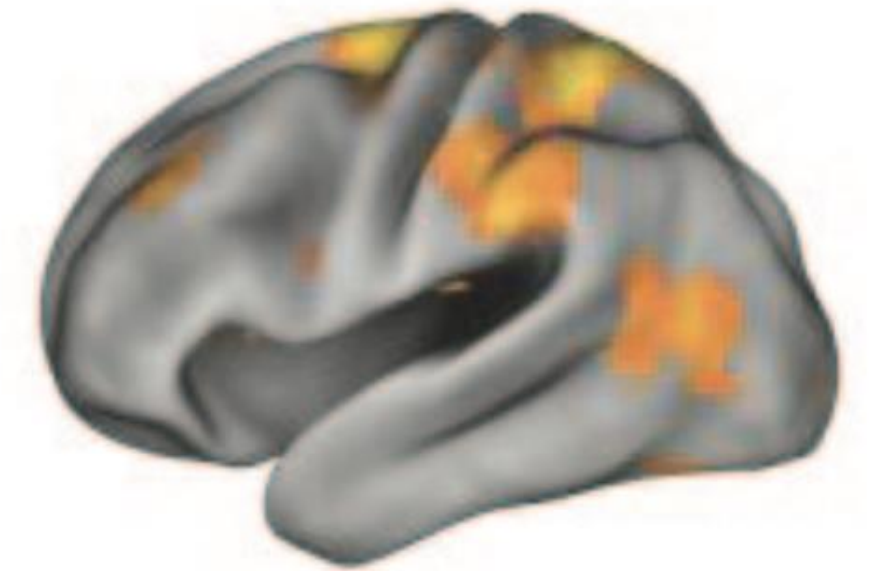
Common illusions & misperceptions

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- Contrast & context skew perceptions of intensity.
- Combining across dimensions can lead to spurious inferences of associations.



Common illusions & misperceptions

- Hue & position lead to perceived distance/difference errors.
- Contrast & context skew perceptions of intensity.
- Combining across dimensions can lead to spurious inferences of associations.
- Humans tend towards categorical biases, even when no categories are present.



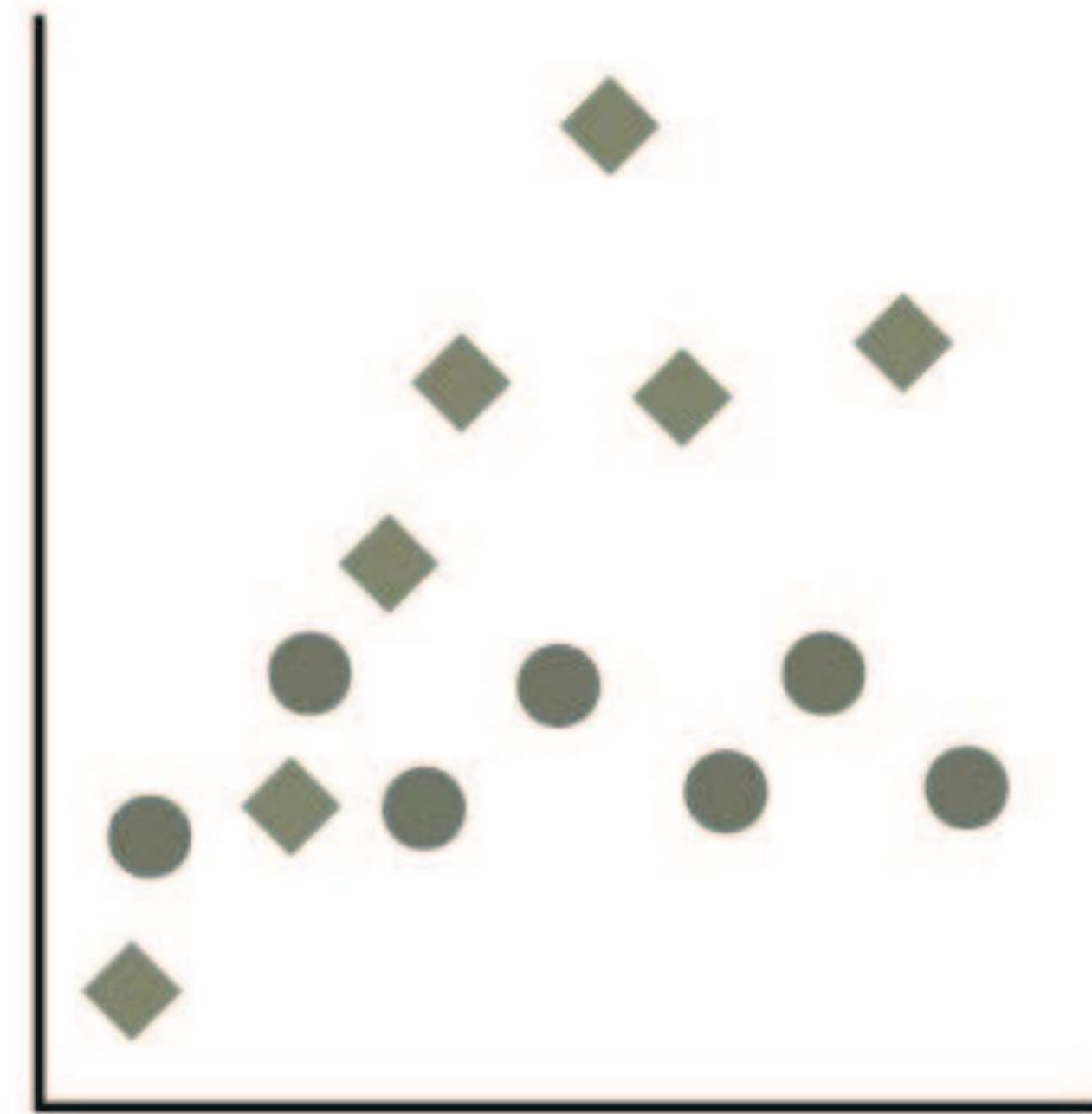
Working with visual biases

Use visualizations to compute fast, simple statistics

Vision quickly extracts simple statistics from data encoded via:

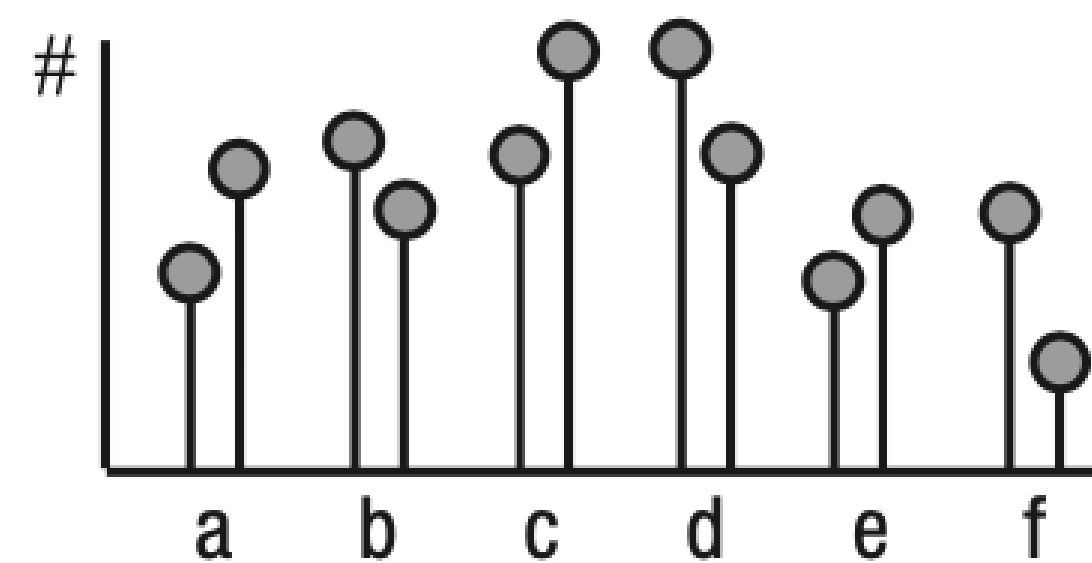
1. position
2. one-dimensional length
3. two-dimensional area
4. angles
5. intensities

Question: Are X and Y associated?

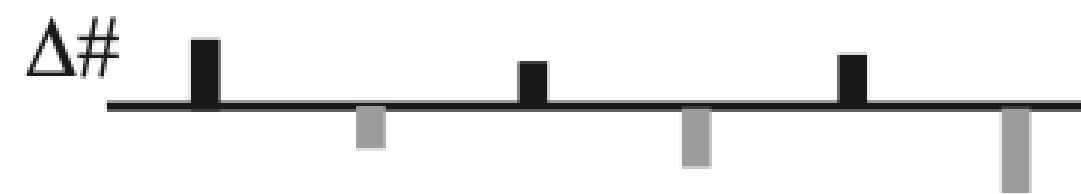


Comparisons as a processing limit

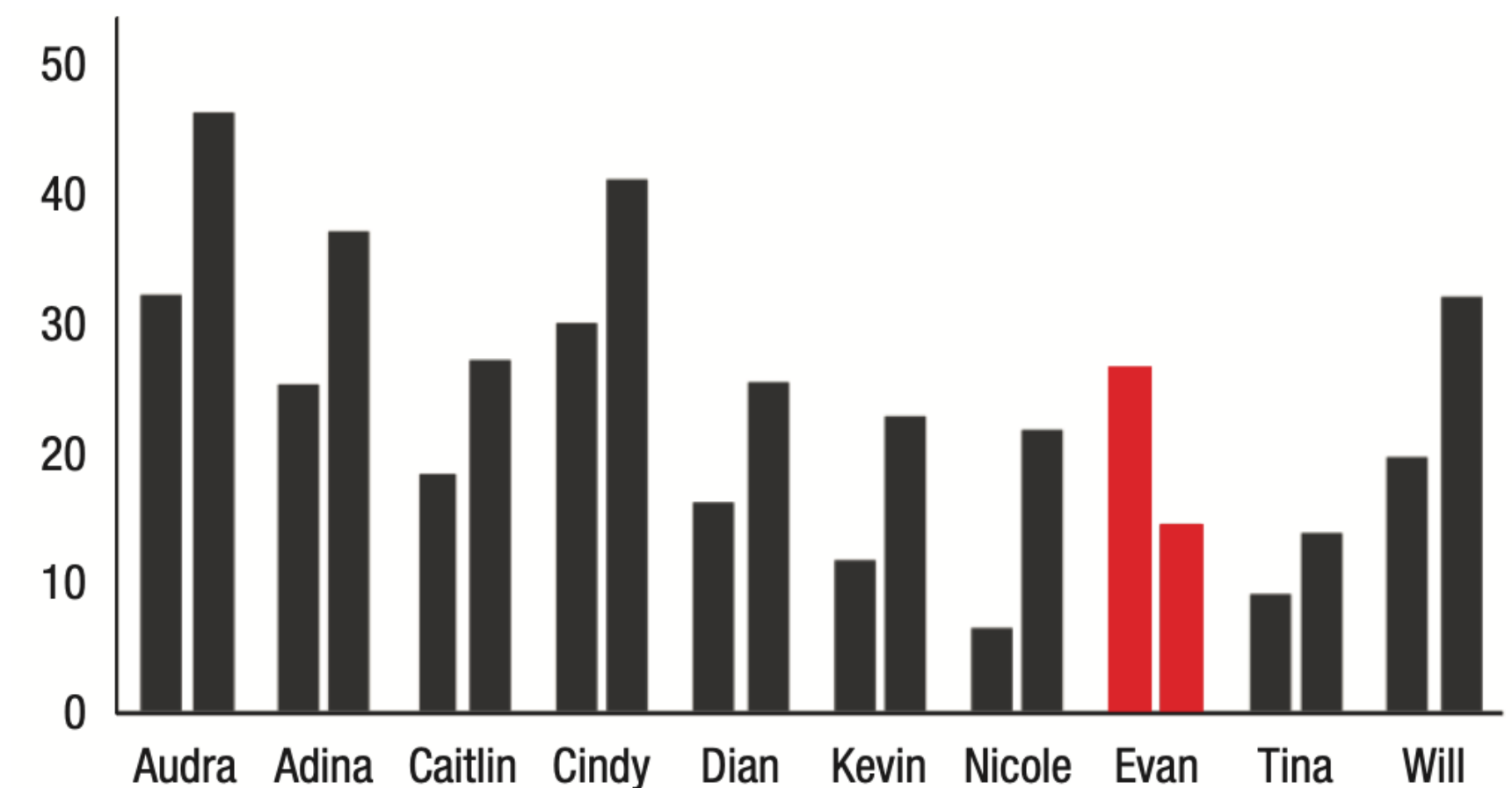
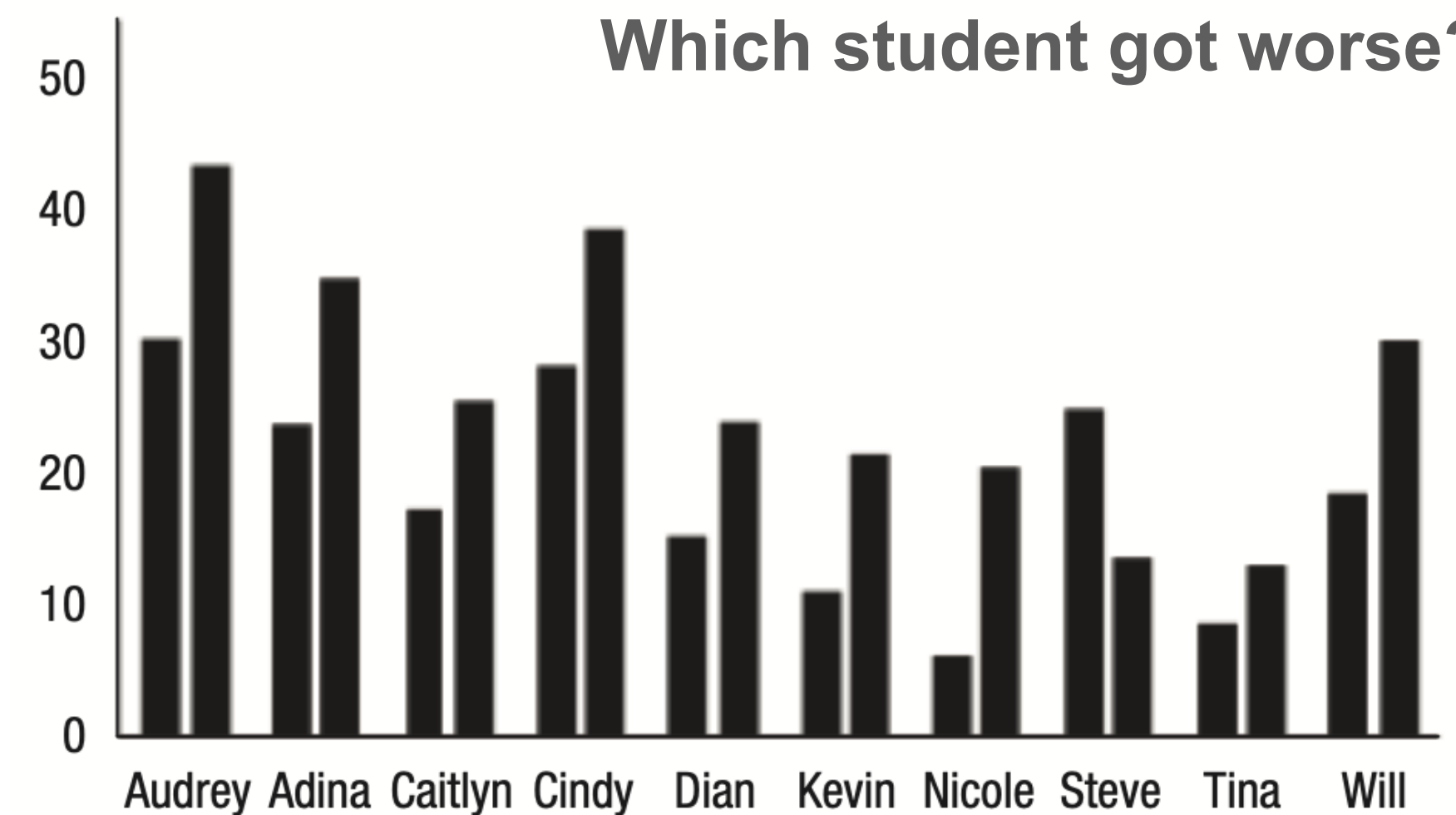
- Seeing general statistical relations (e.g., value, distribution) is *fast*.
- Extracting relationships (and differences) is *slow*.



Tool: Shortcut comparisons by adding direct depictions of the deltas, as below

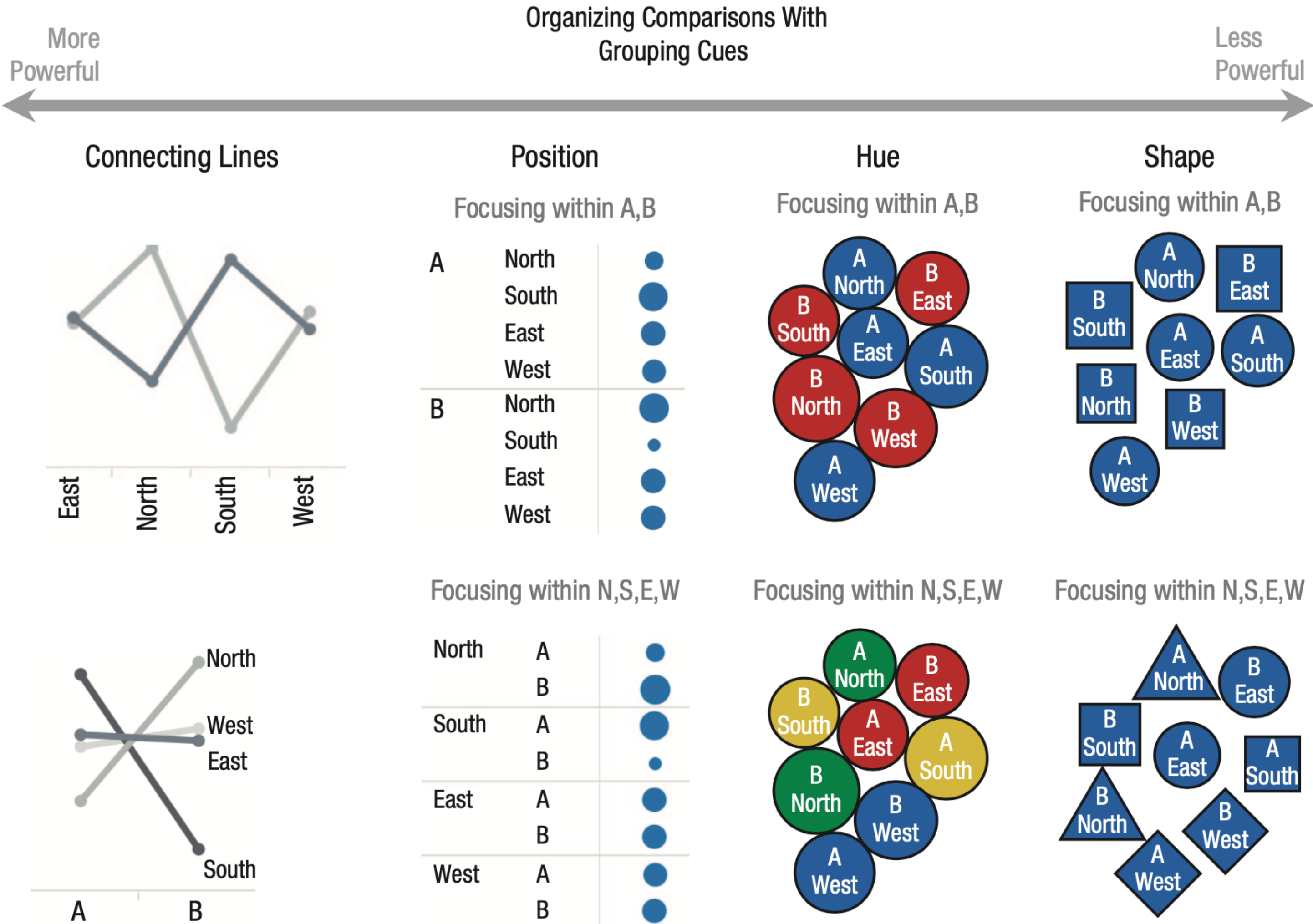


Which student got worse?



Grouping to facilitate comparisons

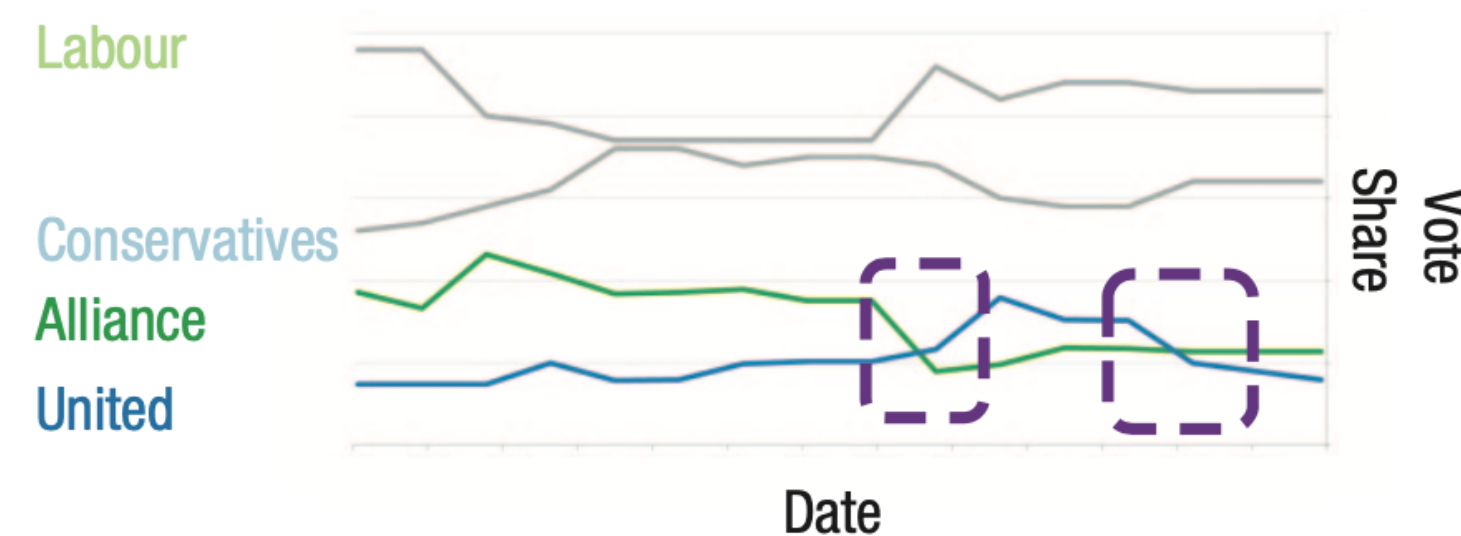
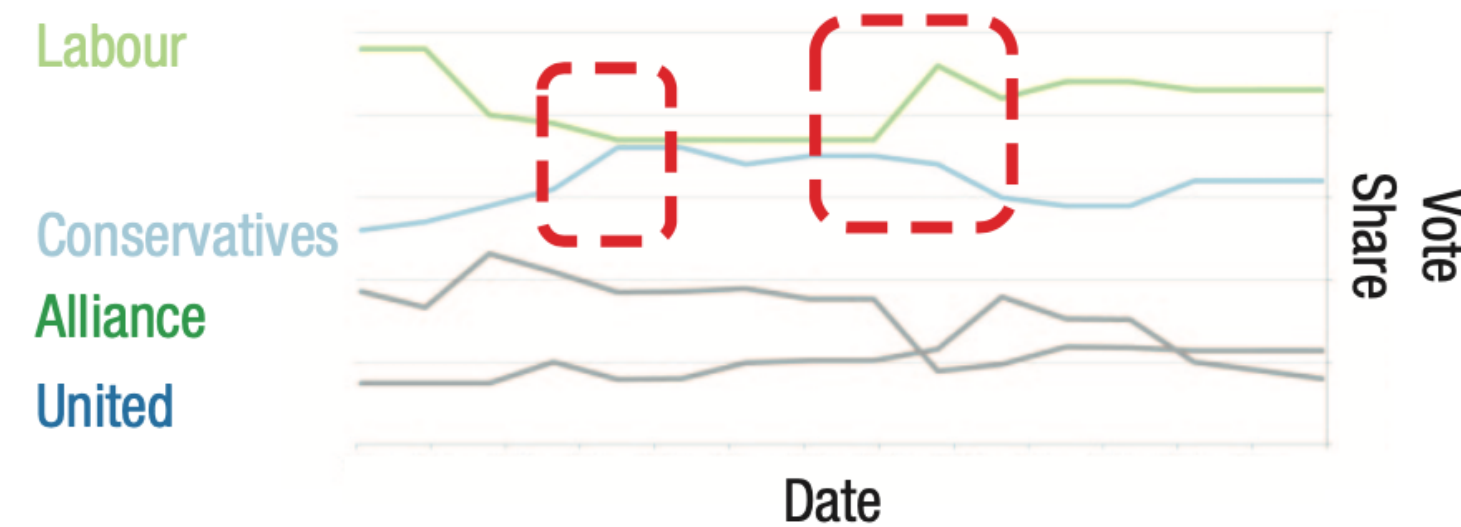
Proximity (position) and hue/shape are natural cues for similarity and grouping respectively.



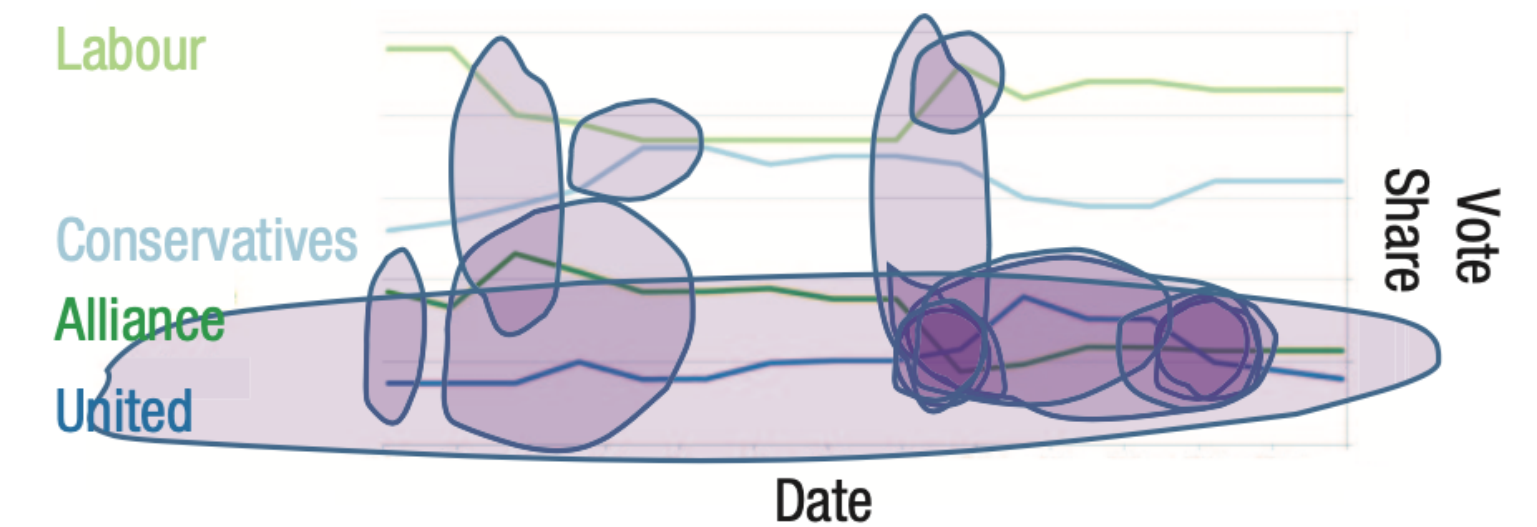
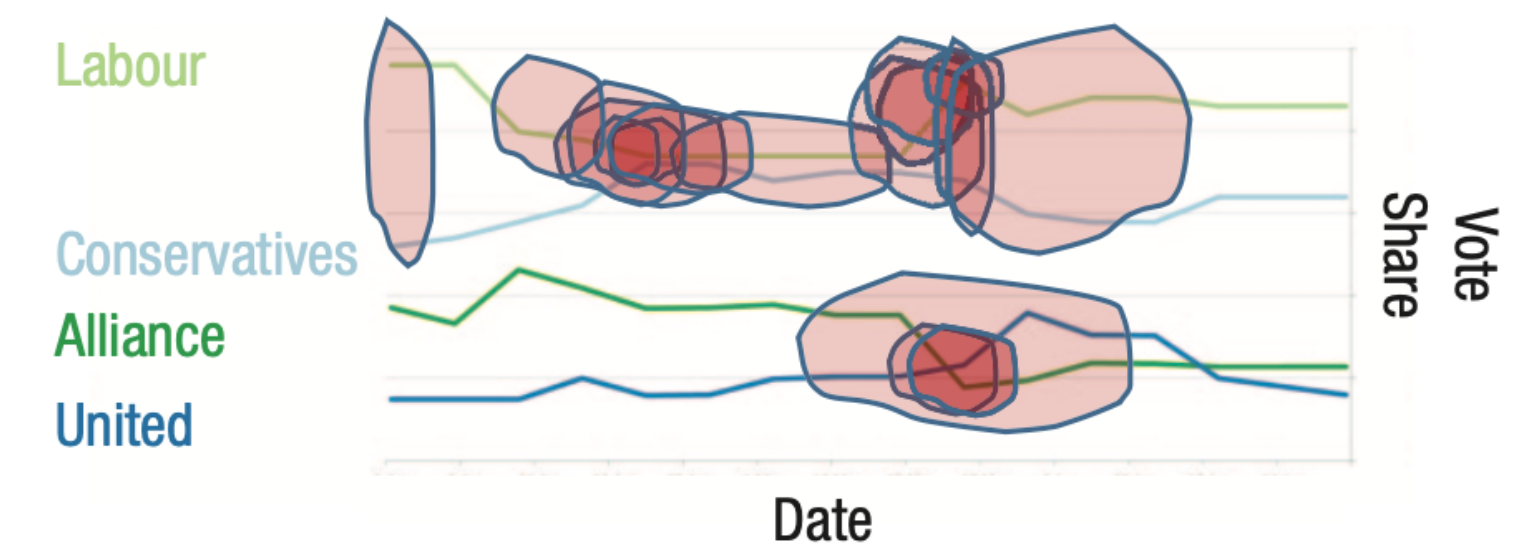
Curse of knowledge

Curse of Knowledge: Inability to simulate the perspective of another viewer.

- Prior knowledge guides how viewer's see graphics.



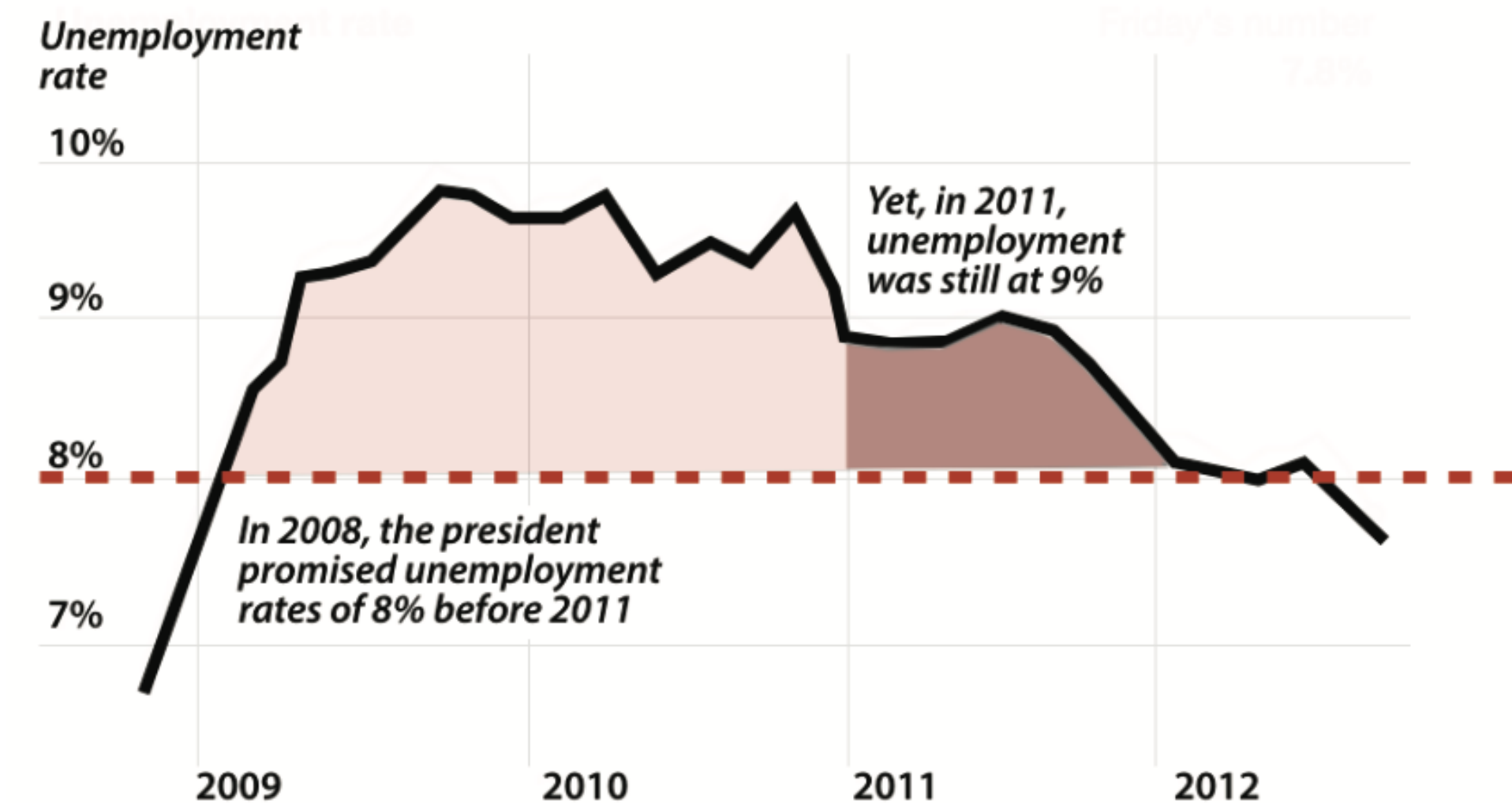
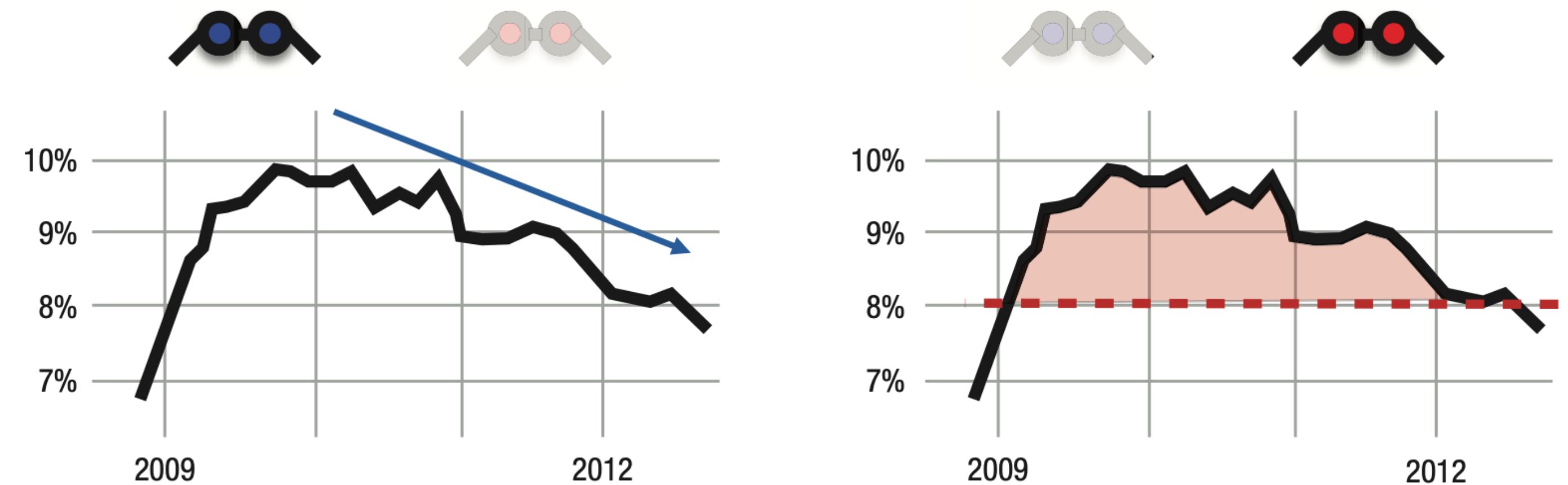
(Xiong et al. 2020)



Curse of knowledge

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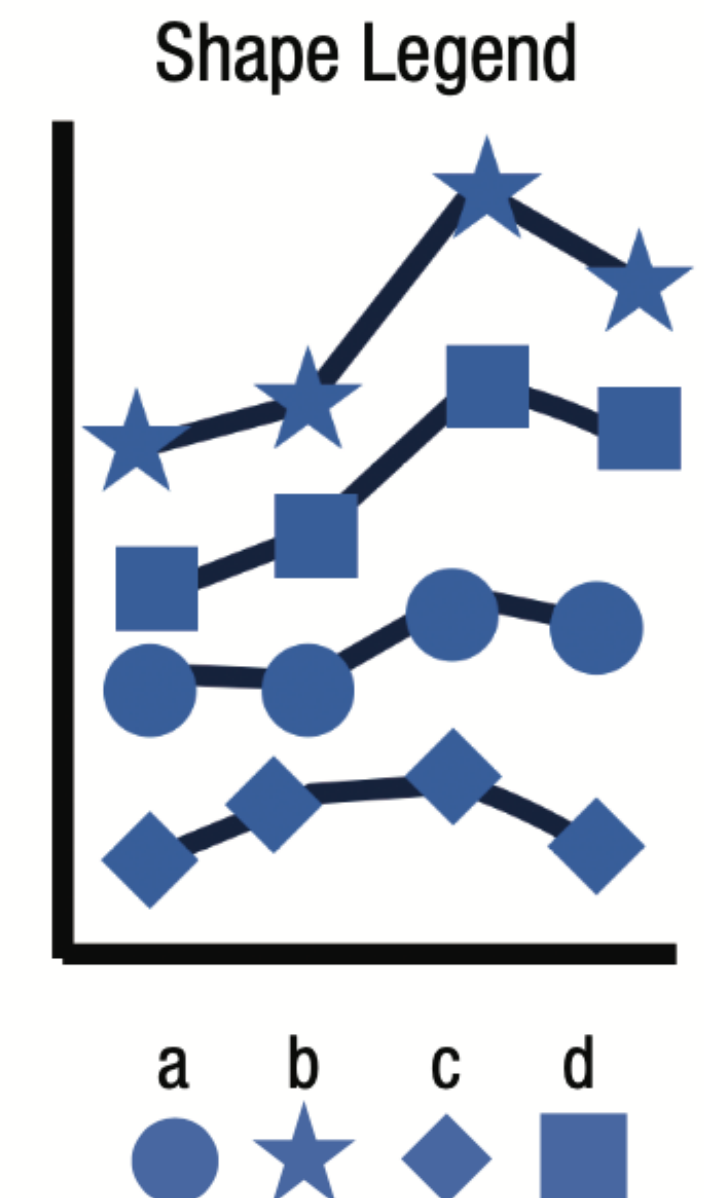
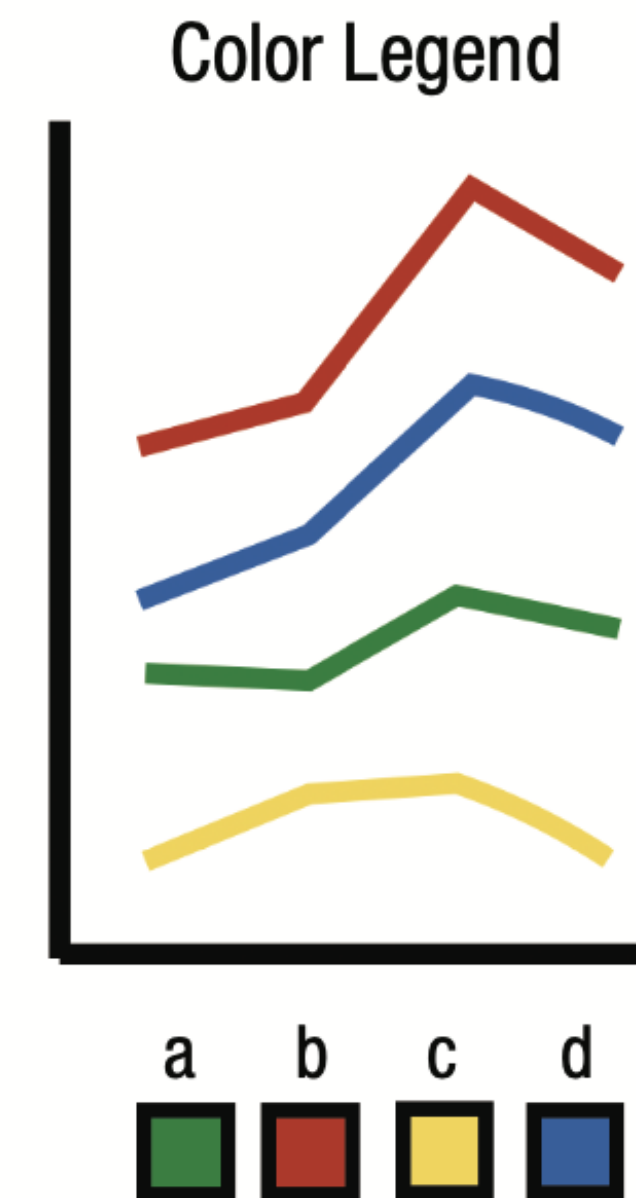
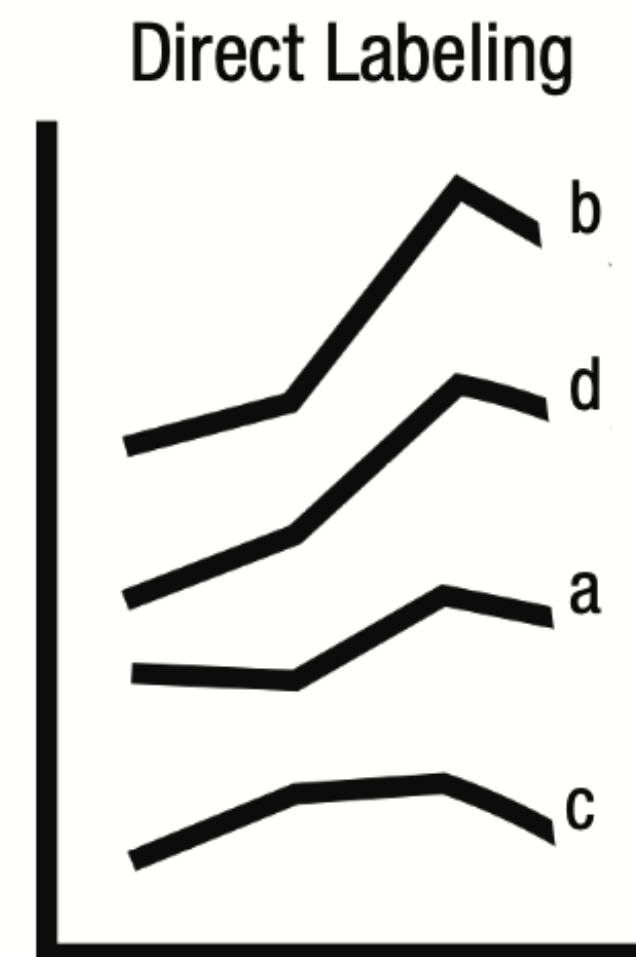
- Prior knowledge guides how viewer's see graphics.
- Use visual features (e.g., color, shading) to guide viewer's eye.
- Use annotations for direct inference.



Inspired by:
<http://www.nytimes.com/interactive/2012/10/05/business/economy/one-report-diverging-perspectives.html>

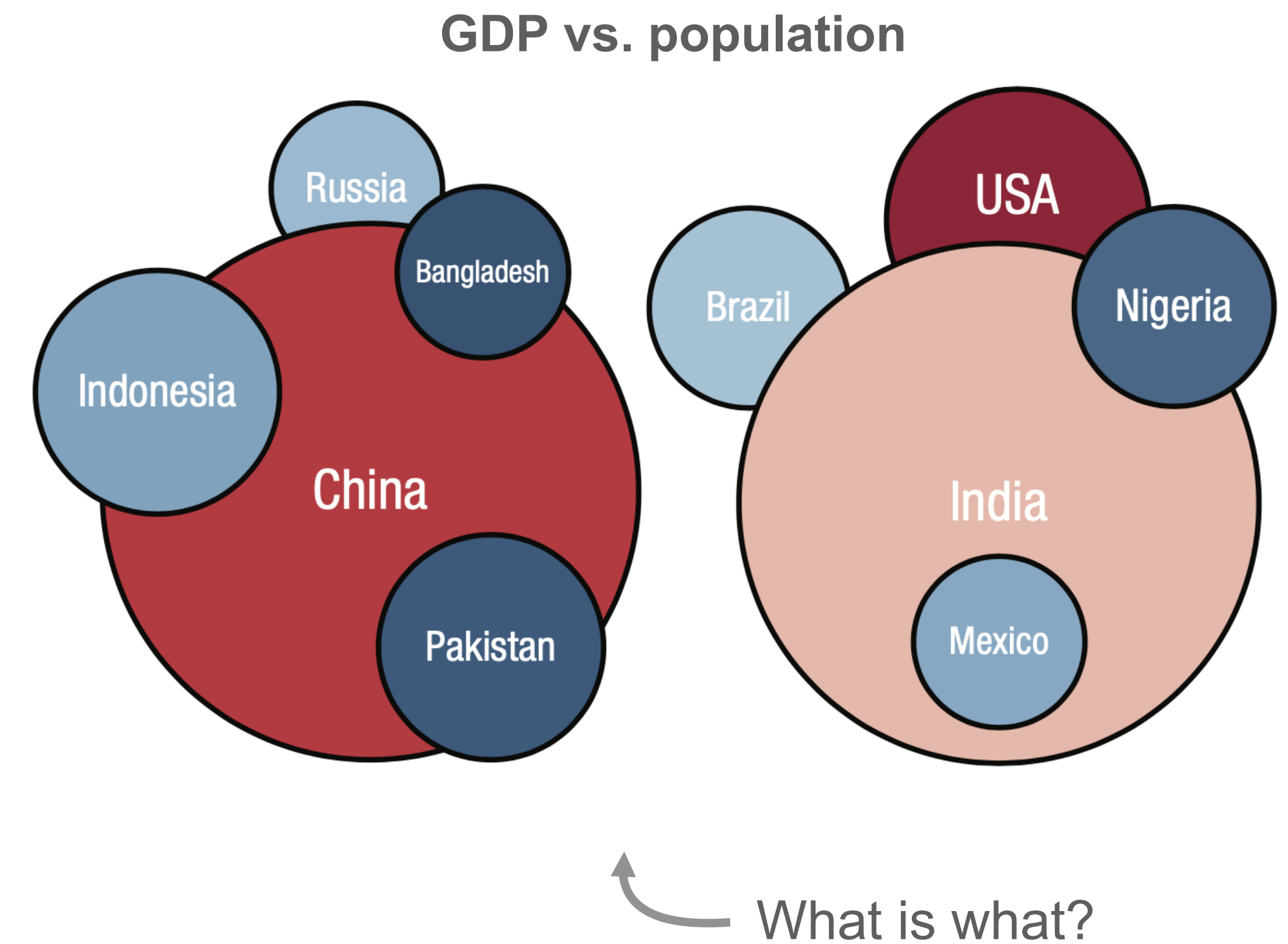
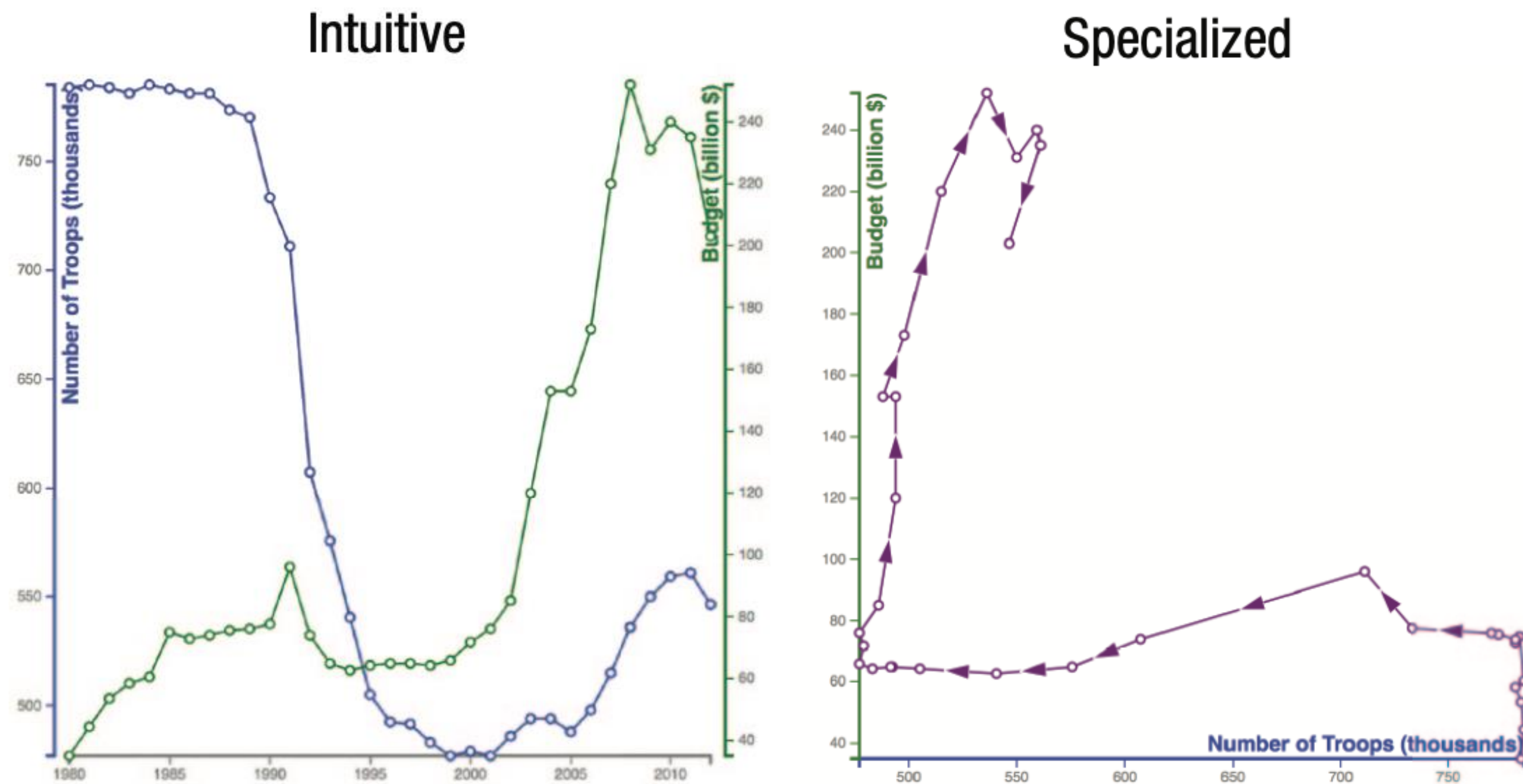
Avoid taxing working memory

People answer questions about data faster and more accurately when data are directly labeled in graph, rather than on legends or captions.



Familiarity facilitates comprehension

Graph schema: A knowledge structure that includes default expectations, rules, and associations that a viewer uses to extract conceptual information from the data visualization.



Respect the viewer's assumptions

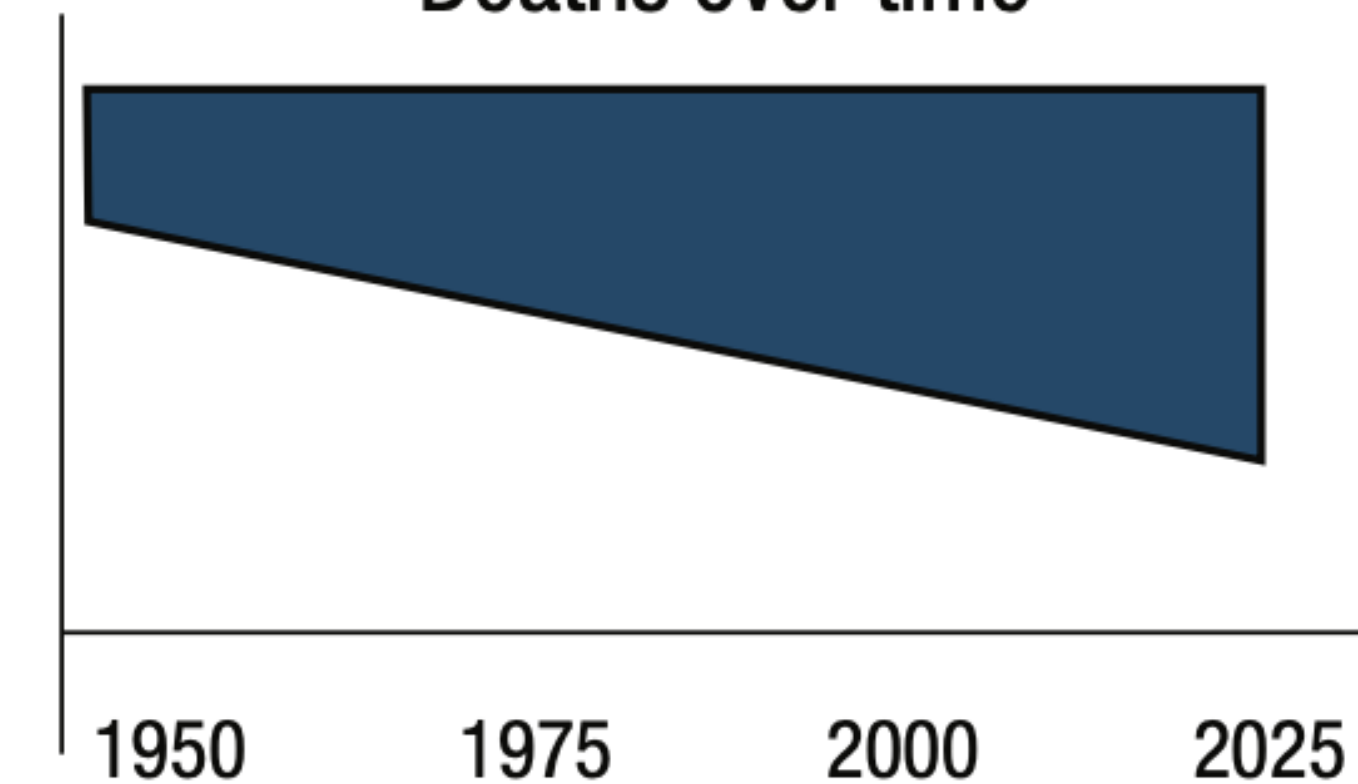
- Some assumptions are "universal"

Length or area $\left\{ \begin{array}{l} \uparrow = \textit{more} \\ \downarrow = \textit{less} \end{array} \right.$

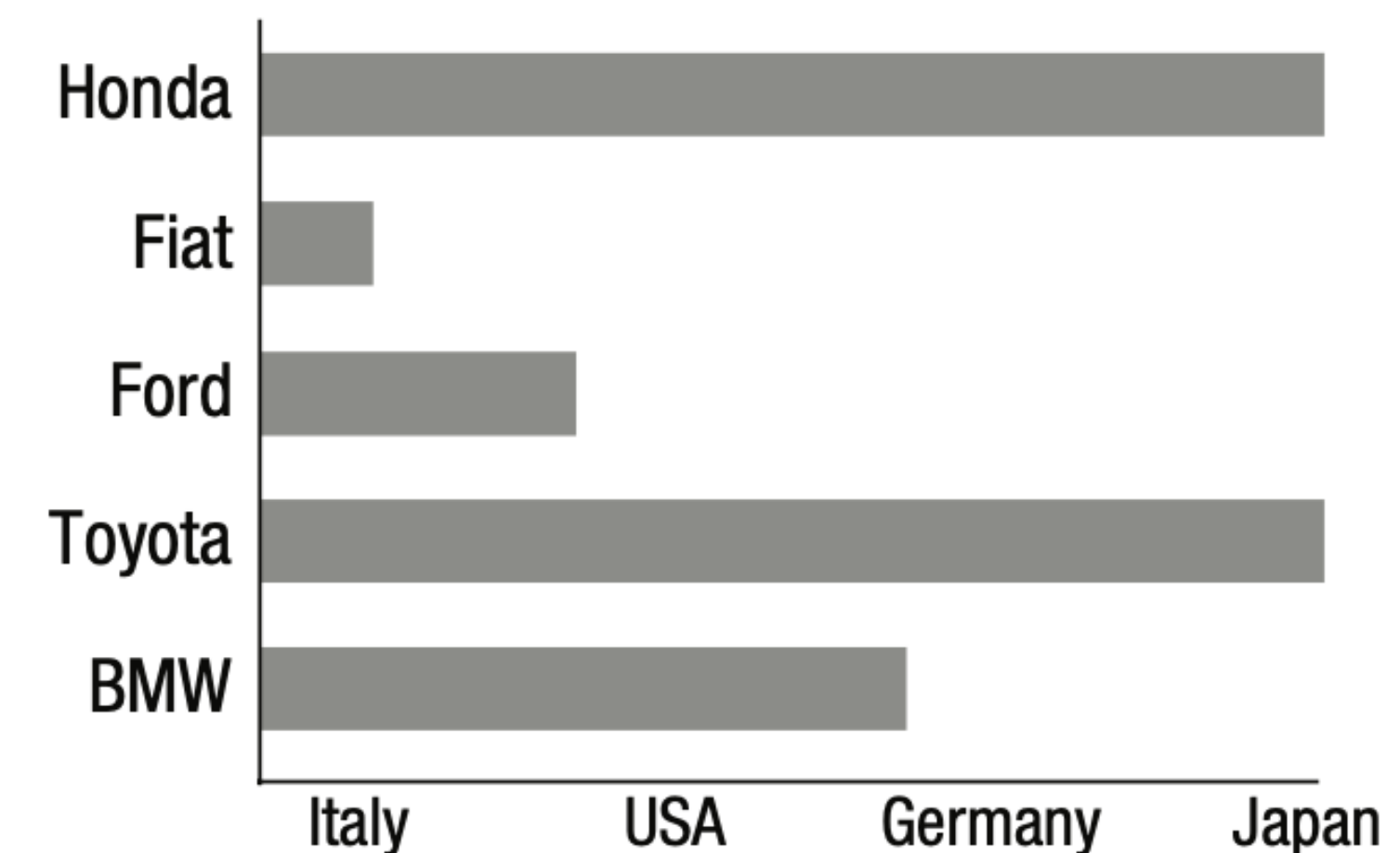
General assumptions

- metric data** → position, length, area, angle, or intensity.
- nominal data** → position, color, shape

Deaths over time

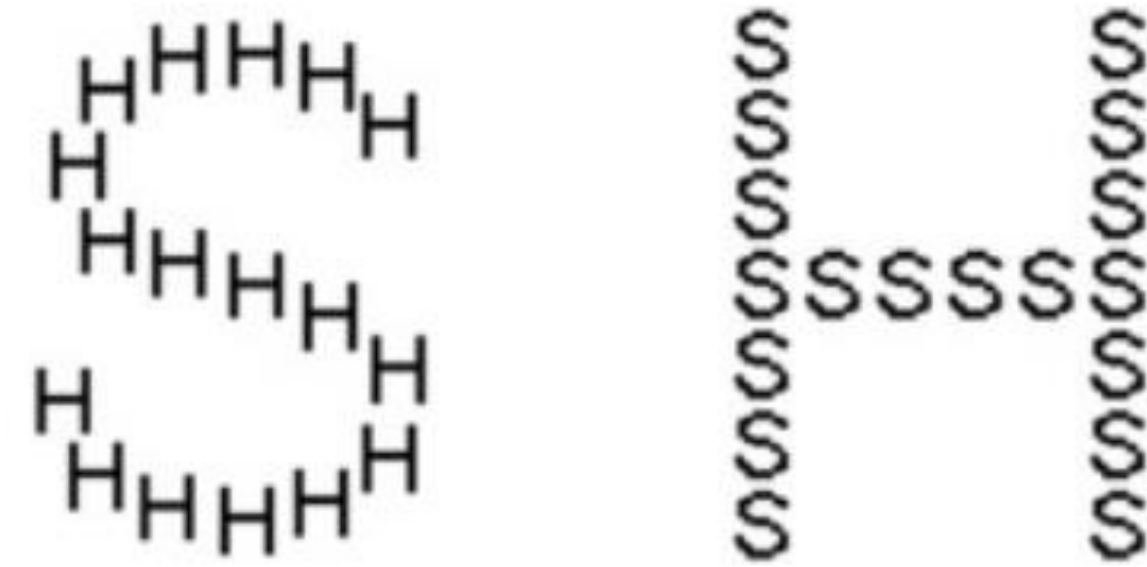


Car maker



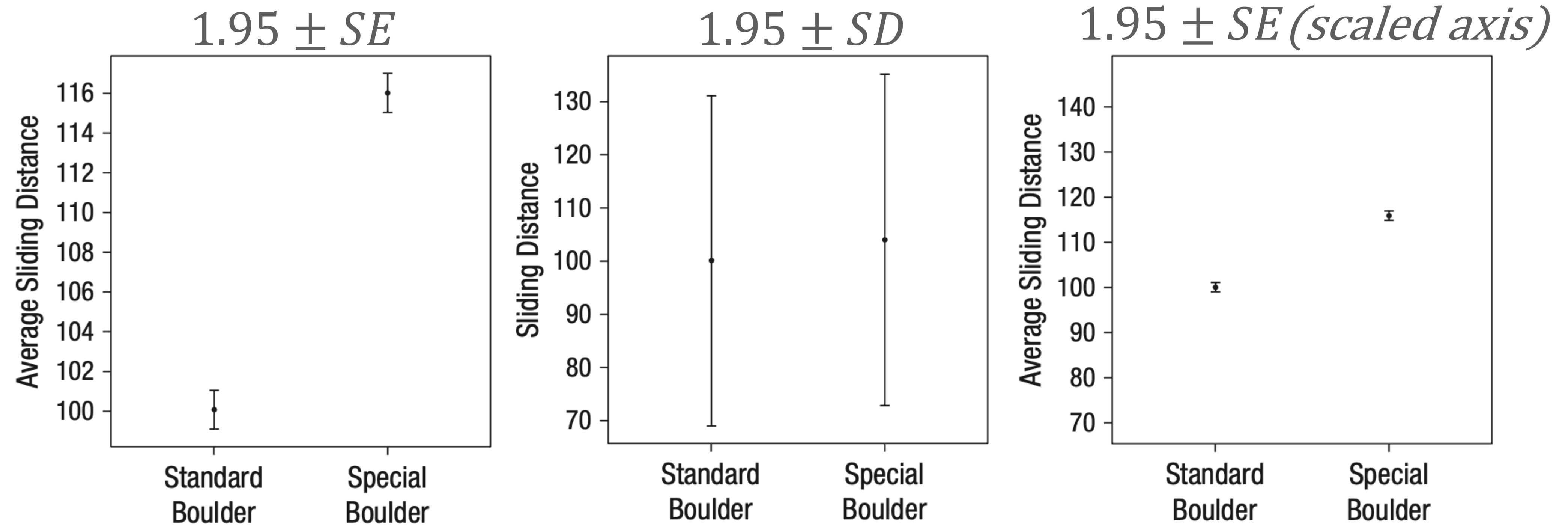
Visual processing → encoding

- **Bottom-up** → saliency
- **Top-down** → features



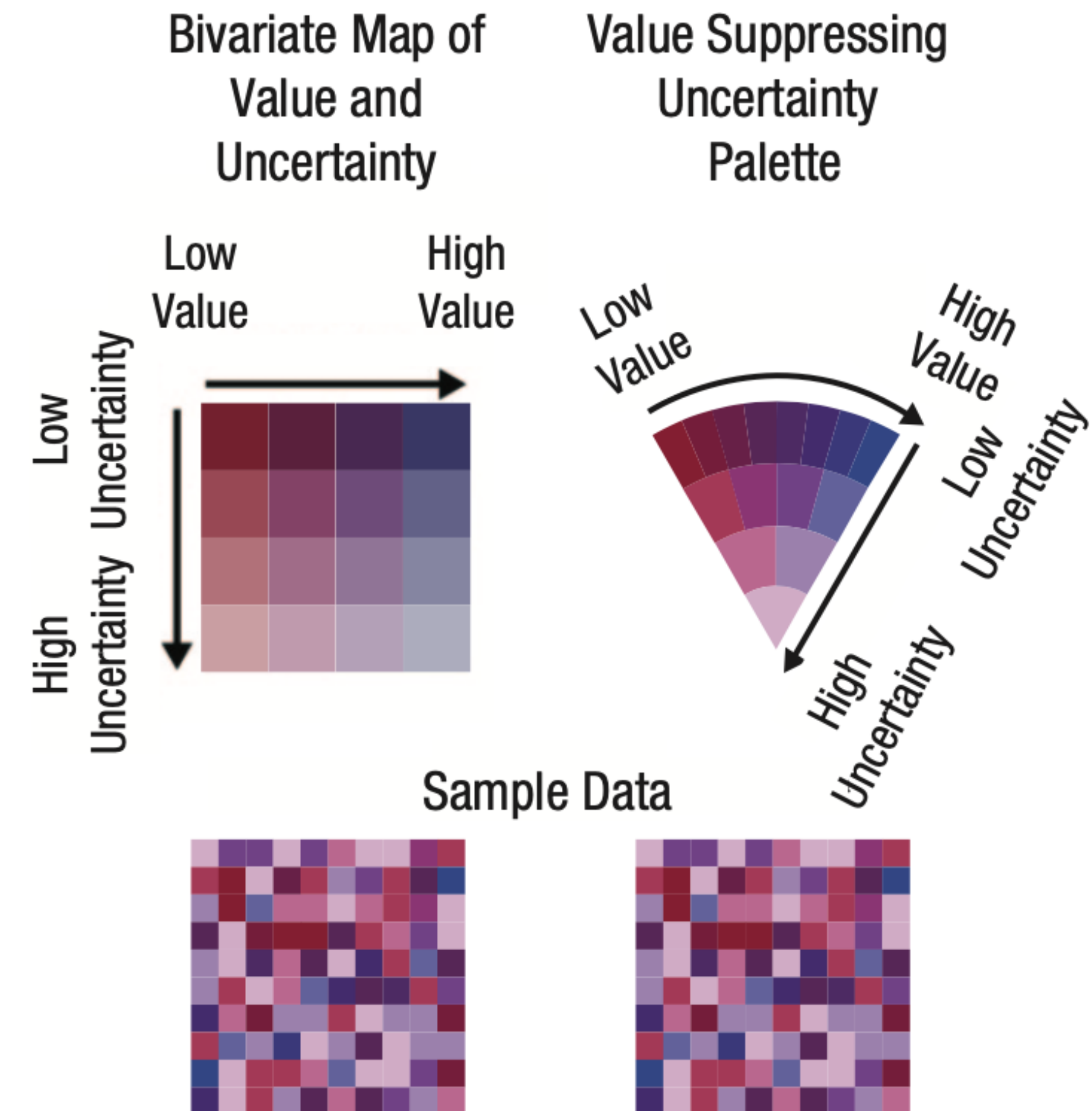
Communicate uncertainty effectively

Errorbars are highly specialized and prone to misunderstanding.



Communicate uncertainty effectively

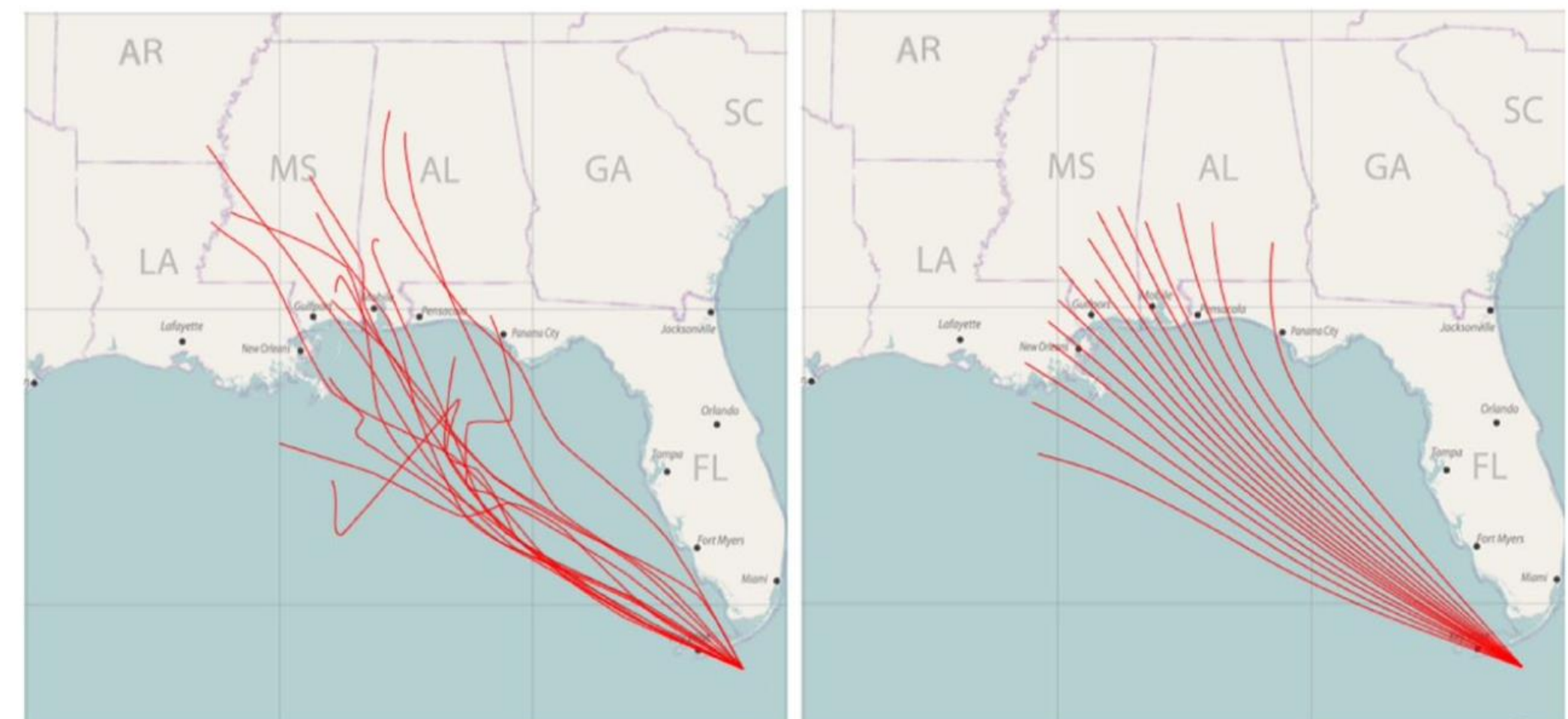
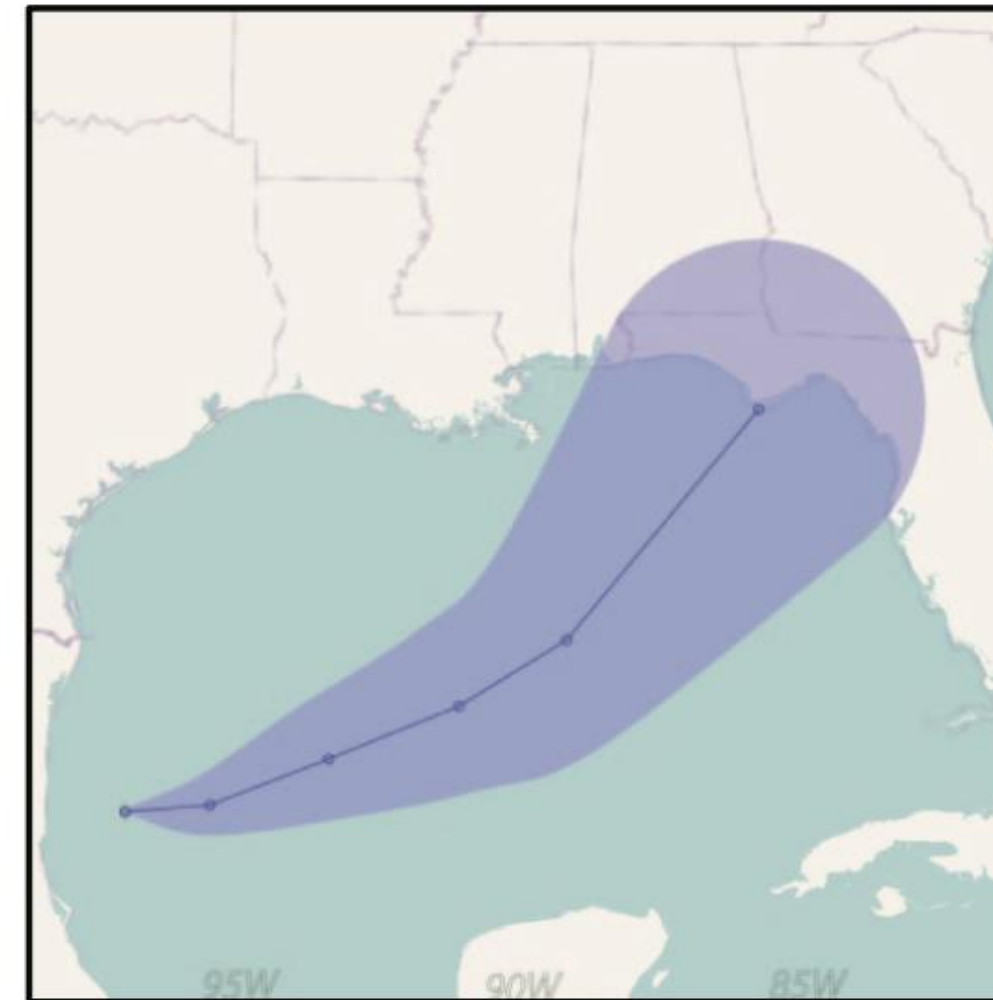
Some visual channels provide subjective impression of uncertainty, but at lower perceptual precision.



The problem of probabilities

Humans struggle with understanding probabilities.

Solution: present probabilities as frequencies or samples



The problem of risk

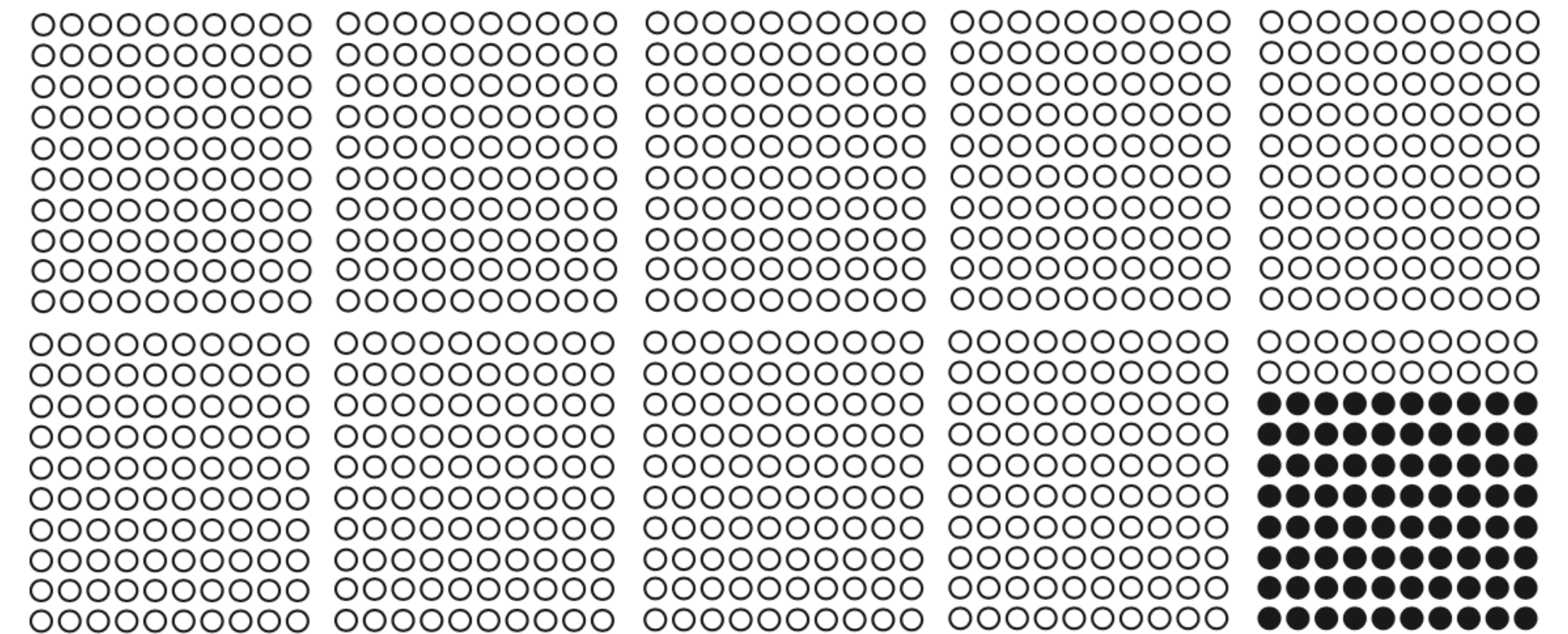
Problems with understanding probabilities lead to problems in understanding risk

- **absolute risk:** overall probability of an experience.
- **relative risk:** comparison of absolute risk between 2 categories or groups.

denominator neglect : people pay more attention to differences in numerators than denominators.

For people with symptoms of arterial disease, aspirin can reduce the risk of having a stroke or heart attack by 13%.

Without aspirin



Guidelines for dataviz for humans

1. Visualize data *before* applying statistics.
2. Beware of visual illusions & misperceptions.
3. Comparisons are slow, global statistics are fast.
4. Avoid taxing working memory.
5. New visualizations must be learned as new schemas.
6. Graph comprehension depends on both bottom-up & top-down visual processing.
7. Communicate your uncertainty effectively.
8. Convey probabilities as frequencies and highlight absolute, not relative, risk.

Take home message

- For the foreseeable future data visualization is used to convey information to humans, so understand human perceptual abilities (and biases) when making decisions about graphical form.